

Poly-Ontological — Unified Math-Clean Specification v1.2

Institute for Mathematical Discovery (IMD)

Unified engineering contracts and mathematical foundations.

0. Global Conventions

Sets. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Vectors/Operators. Hilbert spaces H ; bounded operators $\mathcal{B}(H)$; adjoint $T^\dagger$; spectrum $\sigma(T)$; spectral radius $\rho(T)$.

Norms. $\|\cdot\|$ operator norm; $\|\cdot\|_2$ Euclidean; $\|\cdot\|_1$ L1.

Probability. Random variables uppercase; distributions P, Q .

Categories. Categories $\mathcal{C}, \mathcal{D}$; functors $F: \mathcal{C} \rightarrow \mathcal{D}$; natural transformations $\eta: F \Rightarrow G$; adjunction $F \dashv G$.

Metadata. Every symbol below has a single meaning; units are indicated where relevant.

1. System Overview

IMD consists of:

1. **Engines** — safety-critical services with formal contracts.
 2. **Core Modules** — compact, composable components with fixed I/O.
 3. **Auxiliary Engines** — support services (telemetry, sampling, persistence) with explicit bounds.
 4. **Mathematical Foundations** — poly-categorical semantics and spectral certification used by Engines.
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2. Engines — Contracts

2.1 Interface Schema

For an Engine E :

- **Inputs** $x \in \mathcal{X}$ with unit map $u_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbb{R}^k$.
- **Outputs** $y \in \mathcal{Y}$ with unit map $u_{\mathcal{Y}}: \mathcal{Y} \rightarrow \mathbb{R}^m$.
- **Configuration** $\theta \in \Theta$ (immutable during a run) and **state** $s \in S$.
- **Contract:** predicates
- Safety $\text{Safe}(x, \theta, s)$.

- Liveness $\text{Live}(s)$.
- Performance $\text{Perf}(x, y; \theta) \geq \tau$.

2.2 Runtime Invariants

1. **Boundedness:** $\|y\| \leq b_0 + b_1\|x\|$.
2. **Stability (spectral):** if $A(\theta)$ is the closed-loop operator, require $\rho(A(\theta)) < 1$.
3. **Monotone risk:** empirical risk $\hat{R}_n$ is non-increasing under accepted updates.

2.3 Certification Pipeline

1. **Distributional check (DKW/KS).** For empirical CDF F_n and truth F :

$$\Pr\left(\sup_x |F_n(x) - F(x)| > \epsilon\right) \leq 2e^{-2n\epsilon^2}.$$

At level α , choose $\epsilon = \sqrt{\frac{1}{2n} \ln \frac{2}{\alpha}}$. The classical 5% threshold is $\epsilon \approx 1.36/\sqrt{n}$.

1. **Sequential test (SPRT)** between P_0 and P_1 . With log-likelihood increments $\ell_t = \log \frac{dP_1}{dP_0}(X_t)$, stop at first T with

$$L_T = \sum_{t=1}^T \ell_t \geq \log \frac{1-\beta}{\alpha} \quad \text{accept } P_1, \quad L_T \leq \log \frac{\beta}{1-\alpha} \quad \text{accept } P_0.$$

1. **Spectral certification** of the learned operator W : require $\|W\| \leq r$ and margin γ such that $\sigma_{\max}(W) < 1 - \gamma$.
2. **Fail-safe:** on violation of any invariant, revert to baseline configuration θ_{safe} .

2.4 Acceptance Tests

- **Contract satisfaction:** all three predicates true on a held-out audit set.
- **Calibration:** Brier score $\leq b^*$.
- **Latency:** $\mathbb{E}[T_{\text{resp}}] \leq T^*$; tail bound $\Pr(T_{\text{resp}} > T^{**}) \leq \delta$.
- **Logging:** append-only, cryptographic hash chain.

3. Core Modules — Compact Specs

Each module M exposes $\text{init}(\theta)$, $\text{step}(x, s)$, $\text{reset}()$. Types are total and single-assignment per call.

- **Sampler:** draws IID or martingale-difference sequences with seed discipline.
- **Estimator:** returns $(\hat{\mu}, \hat{\Sigma})$ with concentration certificates.
- **Controller:** computes control $u = Kx$ or $u = \arg \min_u J(u; x, \theta)$; returns closed-loop spectral radius.
- **Certifier:** implements DKW and SPRT; emits pass/fail with proofs and witnesses.
- **Recorder:** immutable logs with digest $h_t = H(h_{t-1} \| m_t)$.

All modules use the units and symbols in §9.

4. Auxiliary Engines — Contracts

- **Telemetry:** bounded overhead $\leq \eta$ fraction of wall time; lossy compression with error $\leq \epsilon$ in ℓ_∞ .
 - **Persistence:** ACID snapshot per certified release; monotone version IDs.
 - **Diagnostics:** property-based tests; counterexamples reduced via delta-debugging.
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5. Poly-Categorical Foundations (PolyCat)

Let $\mathcal{A}$ be a category of **poly-ontological objects** parameterized by a finite **gauge set** $P \subseteq \{\text{primes}\}$. Write the fiber over P as $\mathcal{A}_P$.

5.1 Extension–Restriction Adjunctions

For each $p \in P$, there are functors

$$E_p : \mathcal{A}_P \rightarrow \mathcal{A}_{P \cup \{p\}}, \quad R_p : \mathcal{A}_{P \cup \{p\}} \rightarrow \mathcal{A}_P,$$

with a natural isomorphism

$$\mathrm{Hom}_{\mathcal{A}}(X, R_p Y) \cong \mathrm{Hom}_{\mathcal{A}_P}(E_p X, Y)$$

natural in X, Y . We write $E_p \dashv R_p$.

5.2 Fibration Structure

Let $\pi : \mathcal{A} \rightarrow \mathbf{Idx}$ map objects to their gauge set and morphisms to inclusions. Cartesian lifts exist by the universal property of R_p , giving π a Grothendieck fibration.

5.3 Hubs and Initiality

A **hub** is an object $e \in \mathcal{A}$ with a natural isomorphism

$$\Phi_X : \mathrm{Hom}_{\mathcal{A}}(e, X) \xrightarrow{\cong} eX$$

where eX denotes evaluation of the functor represented by e .

Theorem 5.1 (Initiality of hubs). If Φ_X is natural and eX is a singleton for all X , then e is initial.

Proof. For each X there is exactly one arrow $e \rightarrow X$ corresponding to the unique element of eX ; naturality gives uniqueness.

5.4 Max-Envelope

Fix P and a boundary Hilbert space H_∂ . Let $\rho: \mathcal{A}_P \rightarrow \mathcal{B}(H_\partial)$ be a normal $*$ -representation with ultraweakly closed image. Define the **max-envelope**

$$\overline{\mathcal{A}_P}^{\mathrm{vN}} := \rho(\mathcal{A}_P)^{\prime\prime} \subseteq \mathcal{B}(H_\partial).$$

Theorem 5.2 (Universal normal extension). For any normal $*$ -homomorphism $\phi: \rho(\mathcal{A}_P) \rightarrow \mathcal{B}(K)$ there exists a unique normal $*$ -homomorphism $\tilde{\phi}: \overline{\mathcal{A}_P}^{\mathrm{vN}} \rightarrow \mathcal{B}(K)$ with $\tilde{\phi}|_{\rho(\mathcal{A}_P)} = \phi$.

Sketch. Follows from bicommutant closure and the universal property of ultraweak continuity.

5.5 Rank Bound from Gauge Primes

Let $P_o = P \setminus \{2\}$ and suppose there is a commuting family of self-adjoint idempotents $\{\Pi_p\}_{p \in P_o}$ acting on a finite-dimensional subspace $V \subseteq H_\partial$, with $\Pi_p \Pi_q = \Pi_q \Pi_p$ and $\Pi_p \neq I$. Let the **joint decomposition rank** be the number of nonzero joint eigenspaces.

Proposition 5.3. The joint decomposition rank satisfies $\text{rank} \leq |P_o|$, with equality when the Π_p are linearly independent and mutually sharp on V .

5.6 Size Parameters

For a gauge set P define

$$N = 2^{\prod_{p \in P \setminus \{2\}} p}, \quad L = 256^{\prod_{p \in P \setminus \{2\}} p}.$$

Examples. $P = \{2, 3\} \Rightarrow N = 12288, L = 768$. $P = \{2, 3, 5\} \Rightarrow N = 61440, L = 3840$.

6. SMM Terminal Object

Let **SMM** be the category of structured morphic models used by Engines.

Conjecture 6.1 (Terminality). There exists a terminal object $\top \in \mathbf{SMM}$ such that for all X there is a unique $X \rightarrow \top$ preserving structure. The terminal candidate is the ultraweak closure of the identity representation on H_∂ .

7. Poly-Ontological Formalization

A **poly-ontological mathematical object** is a tuple $(U, \mathcal{S}, P, \mathcal{R})$ where U is a universe of discourse, $\mathcal{S}$ is a signature, P a finite gauge set of primes, and $\mathcal{R}$ relations compatible with P . Morphisms preserve $\mathcal{S}$ and P .

- **Extension/Restriction** coincide with E_p, R_p in §5.1.
- **Foundational claims** are expressed as theorems when supported by proofs; otherwise as conjectures.

Conjecture 7.1 (Foundational initiality). There exists a smallest nontrivial poly-ontological object representing the empty theory over gauge P .

8. Spectral Certification Workflows

1. **Operator construction:** build candidate W from data; enforce $\|W\| \leq r$ by projection.
 2. **Gap analysis:** compute eigenvalues; require a gap $\min_{\lambda \in \sigma(W)} (1 - |\lambda|) \geq \gamma$.
 3. **Empirical validation:** DKW for distributional fit; SPRT for sequential drift; optional martingale tests.
 4. **Release gate:** accept iff all invariants in §2.2 and tests in §2.4 pass.
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9. Symbol and Unit Glossary

- $x \in \mathcal{X}$ input; units depend on sensor domain.
 - $y \in \mathcal{Y}$ output; same dimensionality as task codomain.
 - $\theta \in \Theta$ configuration; dimensionless unless stated.
 - $s \in \mathcal{S}$ state; includes persistent buffers.
 - A closed-loop operator; unitless linear map on normalized state.
 - W learned operator; same space as A .
 - H_θ boundary Hilbert space; inner product units normalized to 1.
 - P gauge set of primes; finite.
 - E_p, R_p extension/restriction functors.
 - Π_p spectral idempotents associated with p .
 - N, L size parameters defined in §5.6.
 - α, β Type-I/II error levels.
 - γ spectral margin.
 - r operator-norm radius.
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10. Implementation Notes

- All modules must be pure functions of their inputs plus explicit state.
- All probabilities and errors are reported with numeric level and sample size.
- Logging is immutable and time-stamped; every certified artifact has a content hash.

- Engines expose a single "certify()" entry point that runs §2.3–§2.4.
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11. Change Log Key (for internal authorship)

- This unified draft harmonizes notation across Engines/Core/Aux and formal sections (§5–§8).
- Open proofs: Conjecture 6.1, Conjecture 7.1.
- Provide concrete module I/O tables per deployment as appendices.